This is the audit log for Fake Coffee Business's data cleaning of the dataset of "Sales Info". It will cover all changes, decisions, and judgement calls made for this dataset.

Initial State

The dataset initially had 8 columns and 10,000 rows. It had missing values and error values like "ERROR" and "UNKNOWN". Every column in the dataset had the exact same type, which was "string". (this will be explained later). The "Transaction ID" column was the only one with no missing values in it.

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
TXN_1961373	Coffee	2	2	4	Credit Card	Takeaway	2023-09-08
TXN_4977031	Cake	4	3	12	Cash	In-store	2023-05-16
TXN_4271903	Cookie	4	1	ERROR	Credit Card	In-store	2023-07-19
TXN_7034554	Salad	2	5	10	UNKNOWN	UNKNOWN	2023-04-27
TXN_3160411	Coffee	2	2	4	Digital Wallet	In-store	2023-06-11
TXN_2602893	Smoothie	5	4	20	Credit Card		2023-03-31
TXN_4433211	UNKNOWN	3	3	9	ERROR	Takeaway	2023-10-06
TXN_6699534	Sandwich	4	4	16	Cash	UNKNOWN	2023-10-28

Column Typing

In a dataset, each column is represented as a "type", depending on what information is stored inside it. For example, a column like "Item" is composed of words, so it is usually assigned the type "string". A column that stores whole numbers may be stored as an "Int" type ("Int" stands for integer). A column storing dates and times may be given the type "datetime", and so on. If a column is given the wrong type, it causes computers to have a harder time processing the information, and makes certain data processing commands harder to do.

Originally, the dataset had "string" for every column type. These were changed to "Int" for Quantity and the type "Float" for price Per Unit and Total Spent. "Float" is the type used for decimal numbers.

Transaction Date was also given the type of "datetime".

Normalization

The columns in the dataset were changed to have underscores instead of spaces in their names. This was done to help the column names be more readable to computers, as sometimes computers interpret names with spaces in them as two different words. Additionally, the whitespace around any words in the dataset was removed. Whitespace is

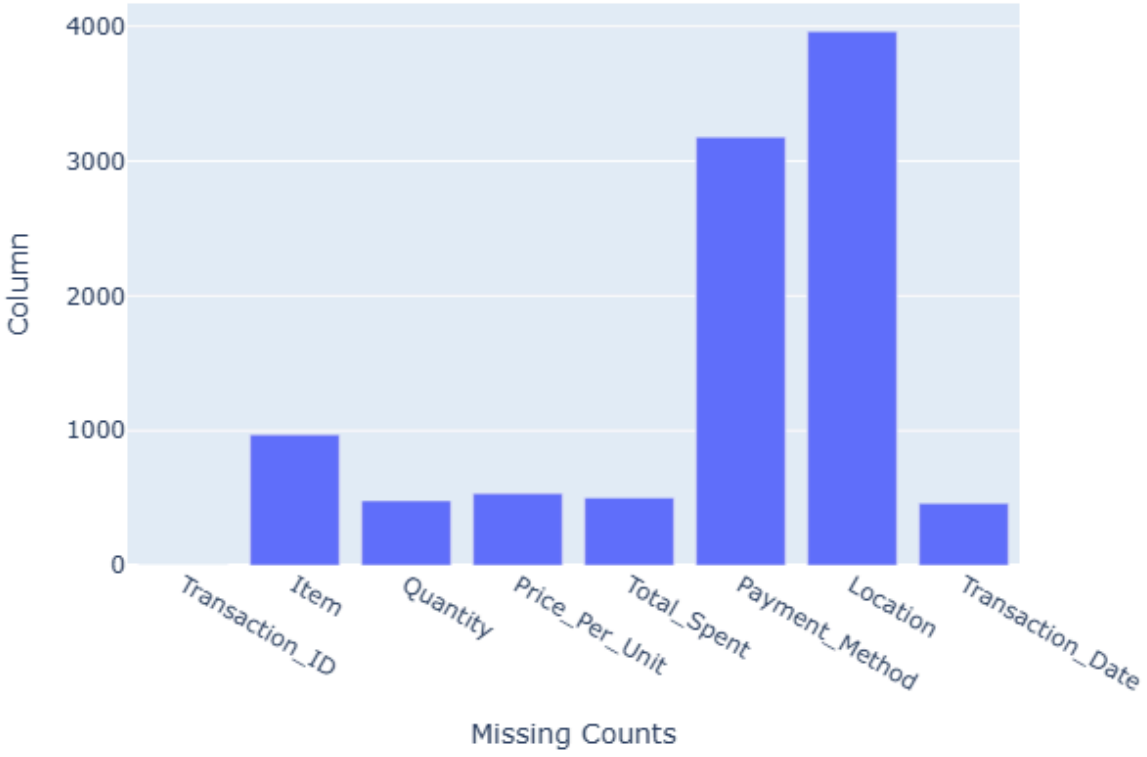
the invisible spaces that may be around certain words in a body of text like this for example.

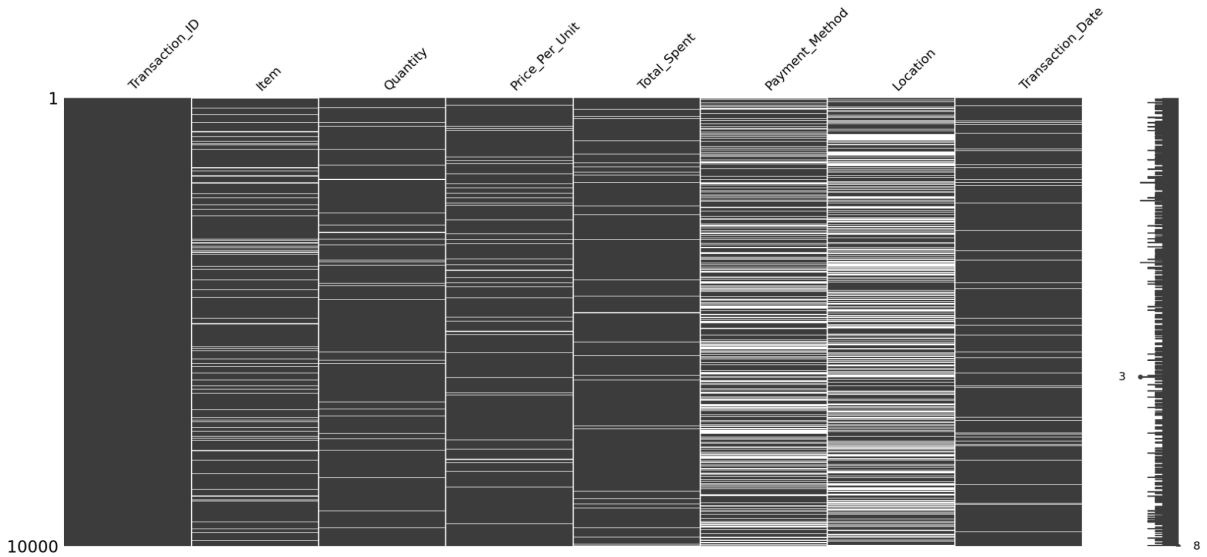
Deduplication

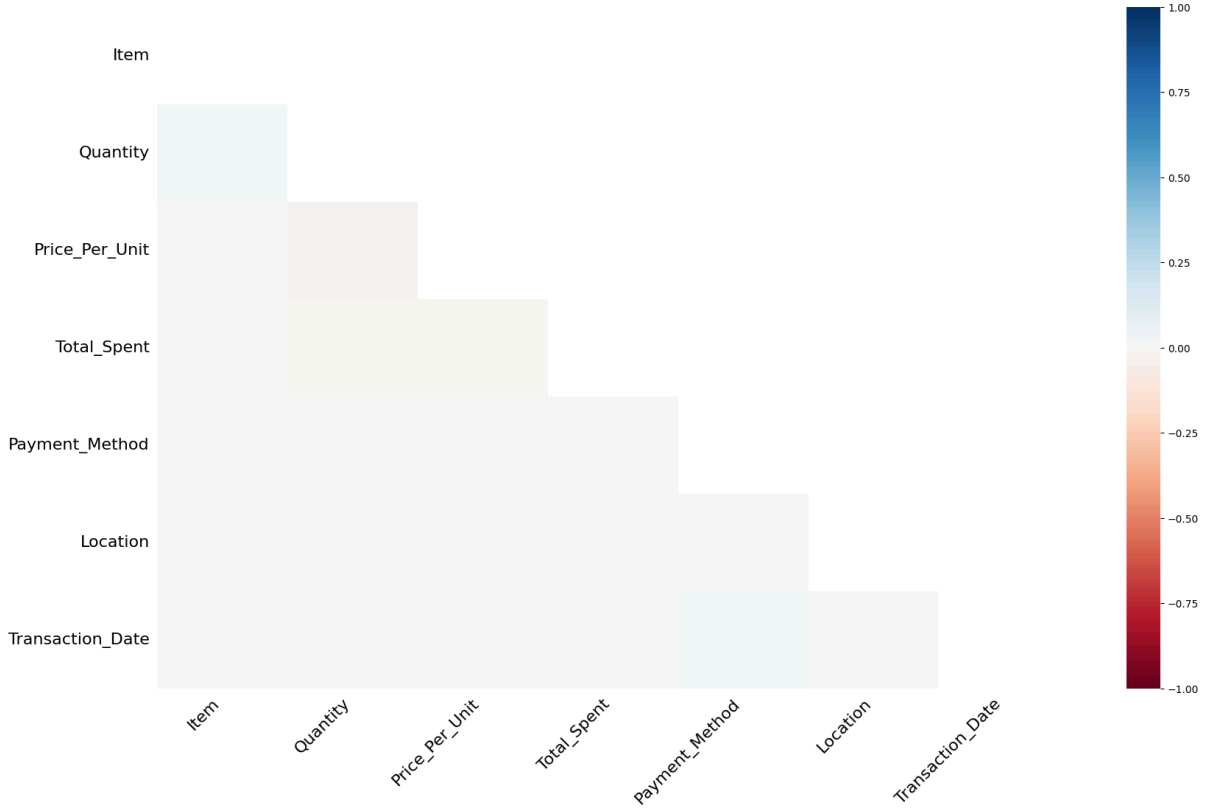
A duplication check was performed. This involves checking how many entries in the dataset were duplicates. Since all of the IDs were different, there were no duplicated entries in the set; no deduplication occurred.

Missing Values

The missing values in the dataset were reviewed. First, a chart was generated that counted the number of missing values per column. This chart was provided to the client alongside all other charts generated during the cleaning of the dataset. A matrix and a heat map was then built to check the relationships between the missing values.







Missing values in a dataset can occur for different reasons. The three biggest categories are MCAR, missing completely at random, MAR, missing at random, and MNAR, missing not at random. There are slight differences between the three, but the only thing the client must know is that this dataset was determined to be MCAR, missing completely at random, meaning there were no special patterns hiding in the missing values. This meant everything was cleaned normally without using any complex statistical techniques.

The first thing to do was to fill the price per unit category. To do this, the code was written to do several calculations based on the existing data to get as close as possible to finding the original price of the item.

If the total spend for a transaction is recorded and the quantity of items purchased is recorded, the total spend was divided by the quantity to get the PPU. If we have the total spend, but not the quantity, we look through the list of possible prices for an item and see if it was possible for a total spend amount with that much money. For example, if the total spend was 33 dollars, we know the transaction was done with either cookies, cake, or juice, because those cost $1 or $3, and 33 is divisible by 3 and 1. We know the transaction was either 33 cookies, 11 cakes, or 11 juices. Once the possible prices are known, we either pick the mode (the most common price in the entire dataset) or we just pick a random number in our set of possible prices. We don't know exactly what got paid for, but this way allows each recorded transaction to at least be feasible. In any other case, like if we only have quantity, or if we don't know anything, we just take the mode. Here are some examples to help understand this process:

Total spend = $14
Quantity = 7

We have both, divide 14/7 = 2. We know this transaction was 7 $2 coffees. PPU = $2

Total Spend = $5
Quantity = ?

We don't have the quantity for this transaction. 5 is divisible by 1 and 5. The mode is 3, so we can't use it here. We take either 1 or 5 randomly.

Total Spend = ?
Quantity = ?

We don't know anything useful about this transaction. Take the PPU as the mode ($3).

Total Spend = ?
Quantity = 22

We know the amount purchased, but we don't know anything else, and the quantity doesn't tell us anything meaningful about the price per unit. Take the PPU as the mode ($3).

Moving now to the ‘Item’ category. The code matches the prices we assigned for the price per unit to the corresponding items. If it detects a PPU of $1, it will assign the missing value as ‘Cookie’, for example. The only two cases where this is different is when the PPU is either 3 or 4: Juice and Cakes both cost $3 and Smoothies and Sandwiches both cost $4. There’s no way of knowing which was purchased, so half of all missing values in the item column with a PPU of $3 were set to ‘Juice’, and the other half was set to ‘Cake’. The same thing was done with the $4 items.

The ‘Quantity’ column was imputed by dividing the total spend by the price per unit. If we don’t have anything for the total spend, we estimate the quantity by taking the mode for the quantity column.

The ‘Total Spend’ column was repaired by multiplying the quantity by the price per unit.

The payment method column was imputed using a method called proportional imputation. The proportions of people paying with a digital wallet, credit card, or cash was about evenly split, so the missing values were put in as one of the three options based on the probability for each option (about 1/3). In simpler terms, since 1/3 of people paid with each method, the missing values were filled with about a 1/3rd being ‘digital wallet’, a 1/3rd being ‘credit card’, and 1/3rd being ‘cash’.

The same method was used for the ‘Location’ column. Note that the proportions being filled were closer to 1/2 since there were only two options.

The ‘Transaction Date’ column was fixed using a different technique called forward filling, where the missing values were given the same date as the date that came before them in the column.

Outliers

The dataset was checked for outliers in the ‘Quantity’ and ‘Total Spent’ columns. Some were found, but these were not deleted or altered, as the data looked feasible. The only noteworthy thing here was that all outliers in the total spend category were people purchasing 5 salads for a total of $25. No direct explanation for this exists, but the client is encouraged to investigate further why this may be the case.

Final Checks

The dataset was double checked over, then was exported as a csv and turned back into an excel file. A table was made in the excel file for easy viewing. All additional files created were put into the delivery file among the original file, cleaned csv, and excel table.

Limitations and Judgement Calls

The amount of missing items in the ‘Payment Method’ and ‘Location’ columns was very high, almost high enough to be scrapped from the clean dataset. Client should be advised these columns may not accurately capture the real numbers for these metrics. The client is suggested to investigate and repair any issues causing the missing values to not be recorded properly, especially considering how important having accurate financial records are for taxation purposes.

The transaction date column had the lowest amount of missing values, but client should be advised that trends over time may not be completely accurate, especially for large amounts of purchases that are recorded to have happened on the same day.